

# EMCal calibration update: data/sim comparisons

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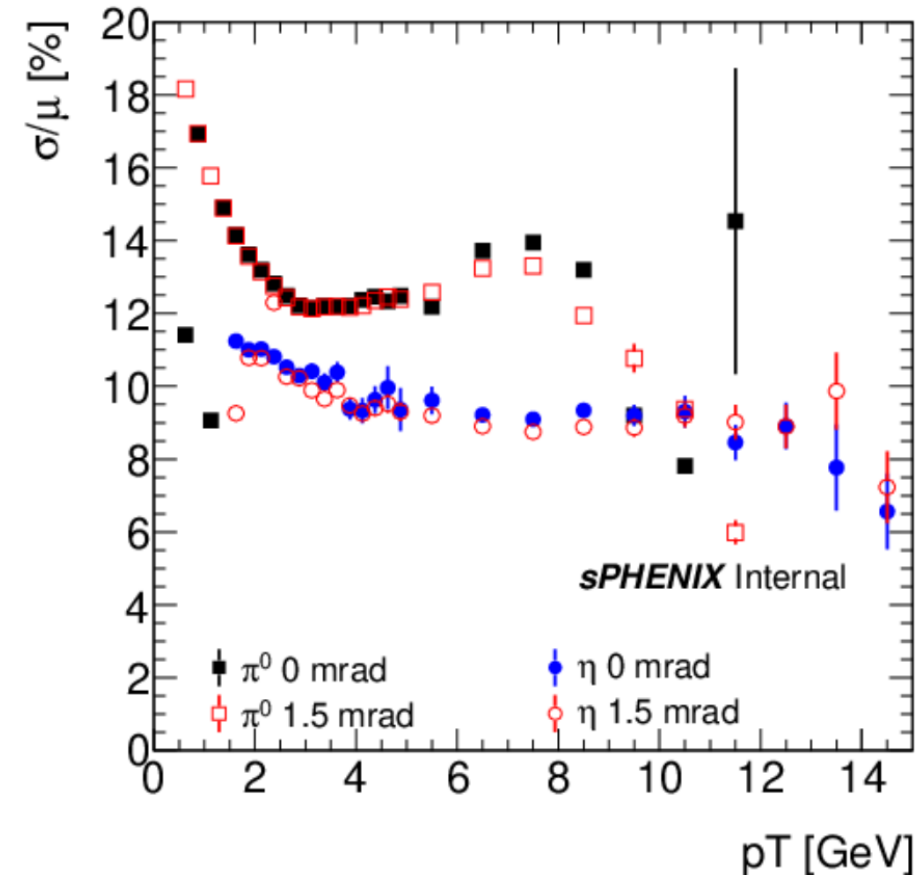
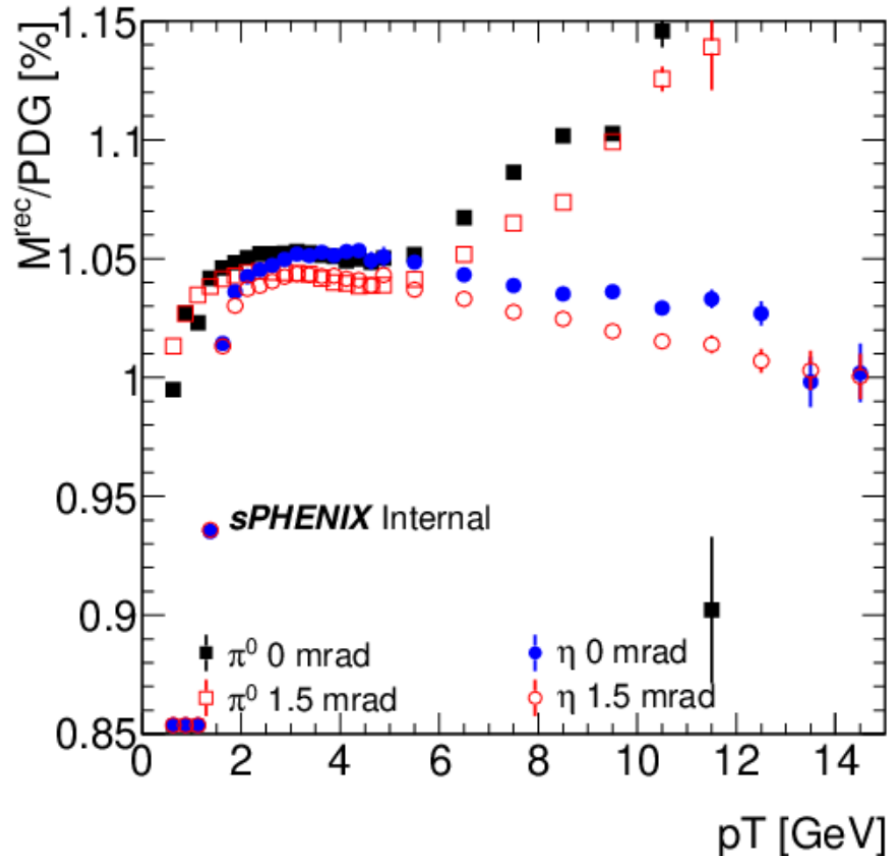
# Selections

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- **Data: ana509, GRL**
- **Pythia run 28: MB, JetXX, Jet12+MB (double interactions)**
- **MBD vertex  $|z| < 20$  cm**
- **Cluster selections**
  - $p_{T1,2} > 0.25$  GeV
  - $Asym < 0.6$

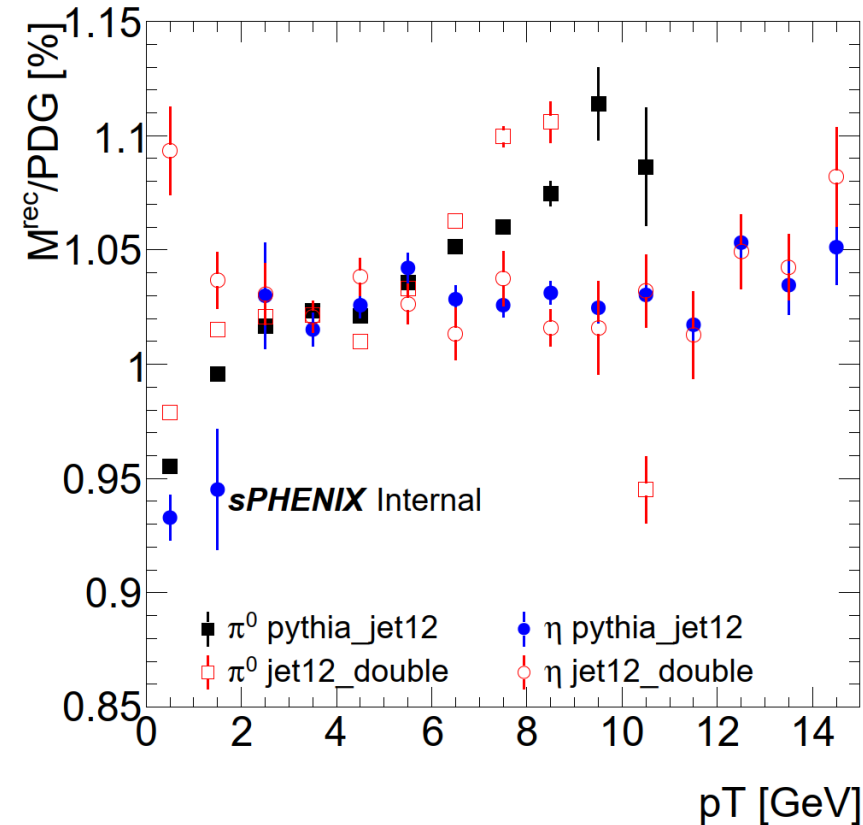
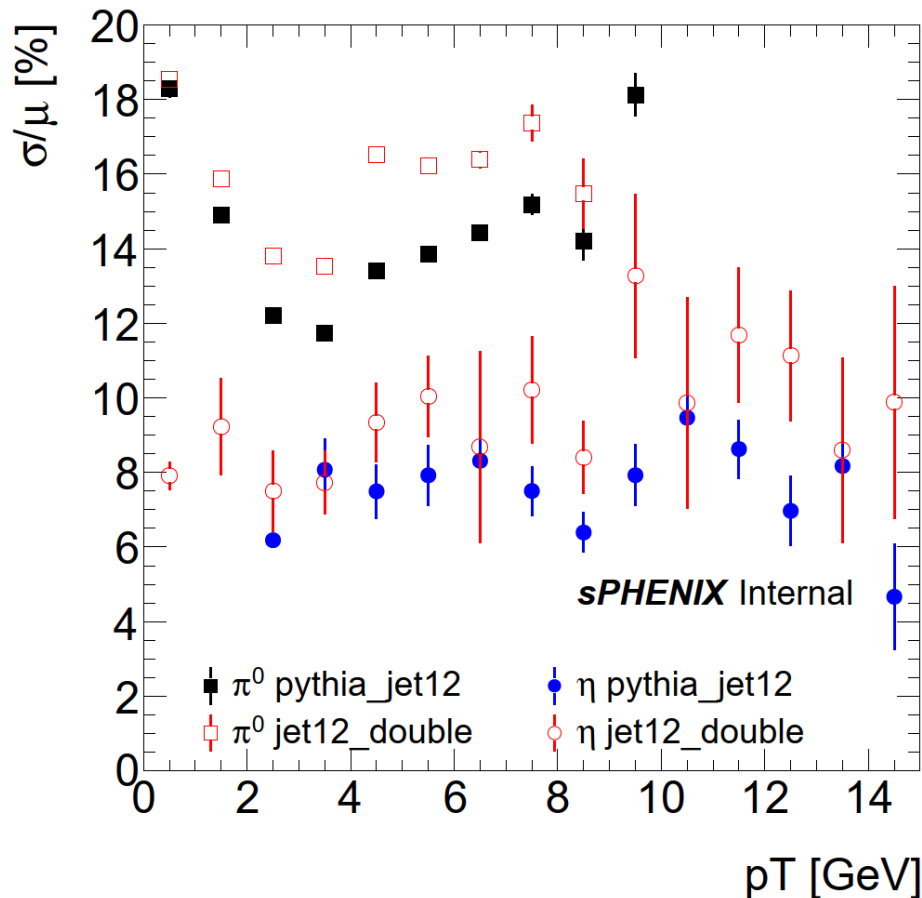
# 0mrad vs. 1.5 mrad data

- 0mrad (1.5mrad) is estimated to have ~22% (7%) double interactions, which degrades vertex resolution
- This degradation is worse in 0mrad because the vertex width is ~50 cm where it is ~15cm in 1.5mrad
- For the rest of this presentation use 1.5 mrad for data/MC comparisons



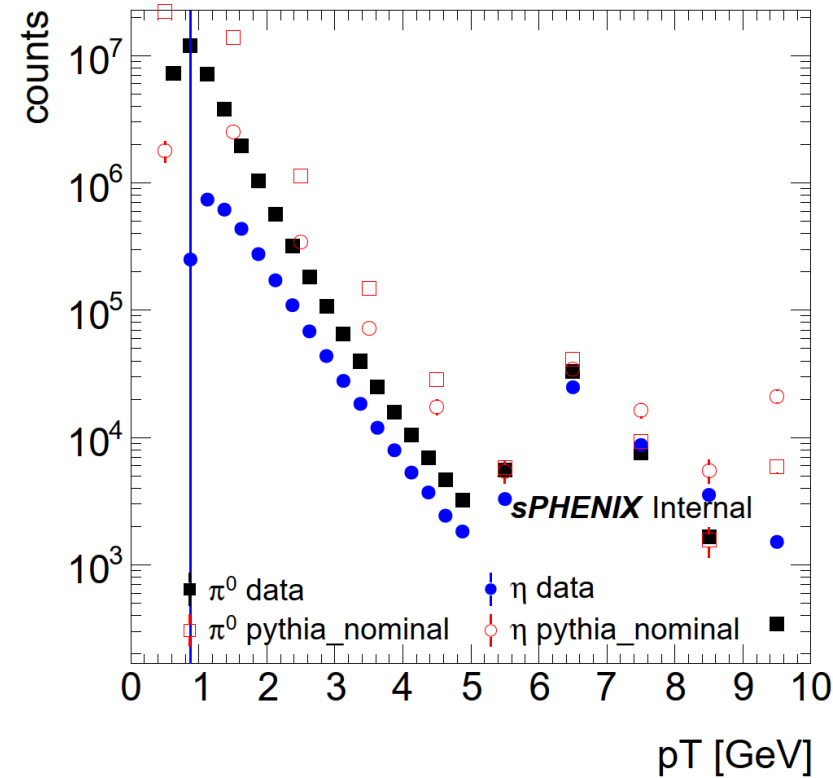
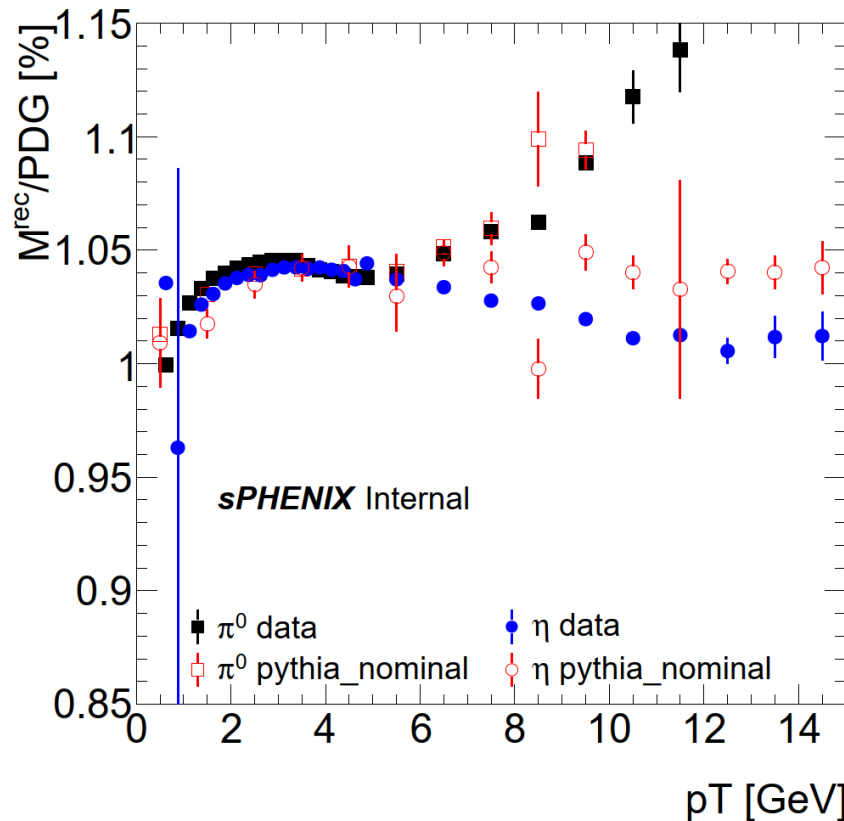
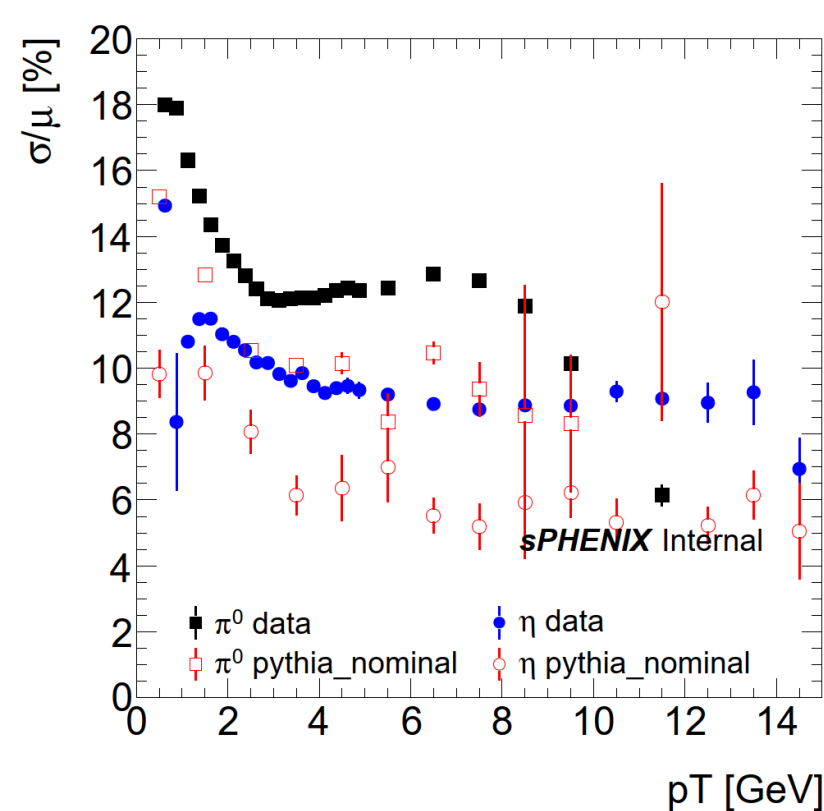
# Pileup effects in MC

- Large effect seen in double interaction sim
- Could be effect of having the wrong vertex or could be poor performance at large truth z vertex which enters the because of a bad reco that is less than 20



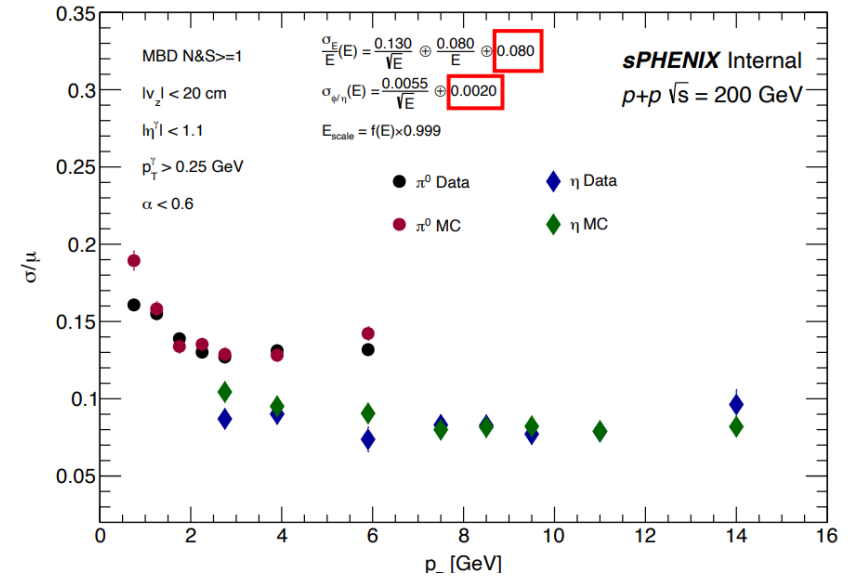
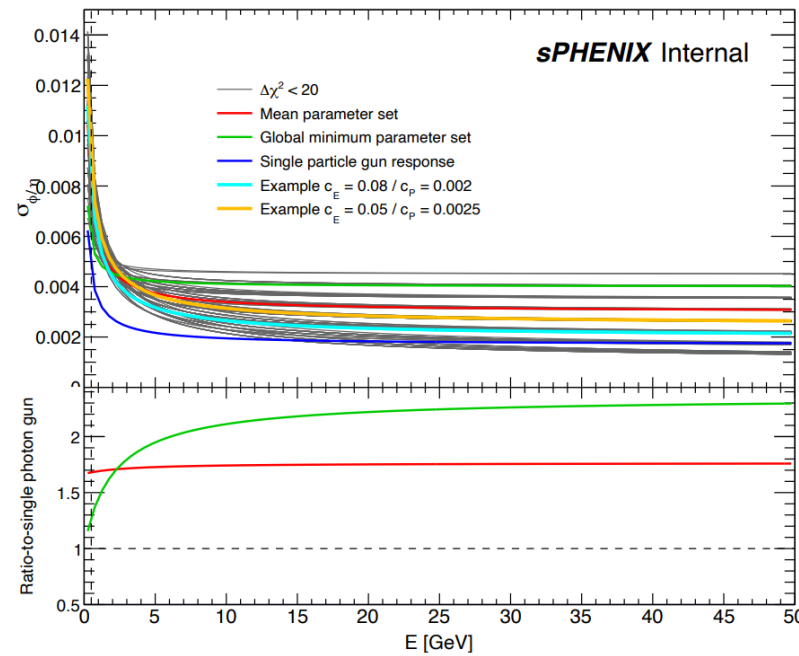
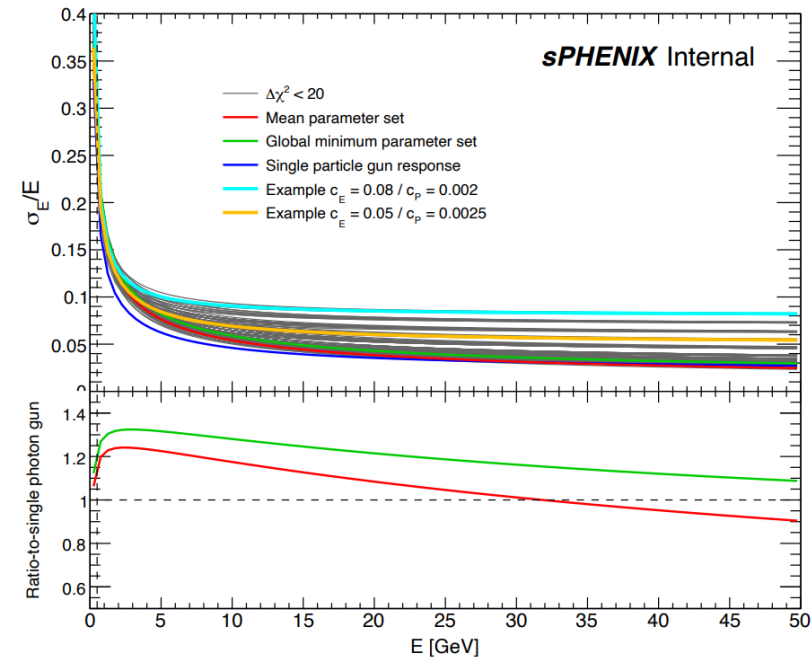
# Data pythia comparison

- Pythia: MB + jet5 ( $p_T > 6$  GeV) + jet8 ( $p_T > 9$  GeV) + jet12 ( $p_T > 12$  GeV)
- Data is much wider than sim...
- It is unclear why, and thus we are attempting to smear the MC



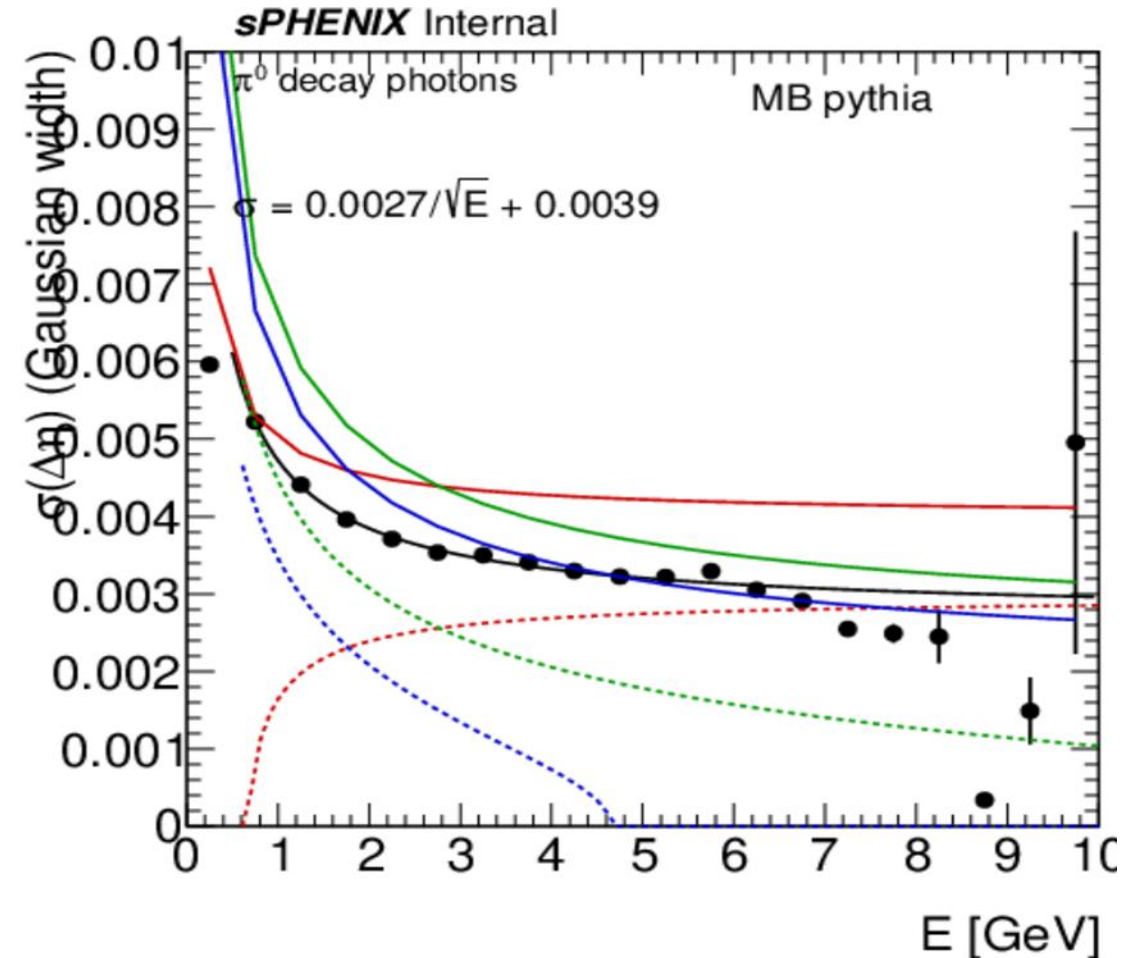
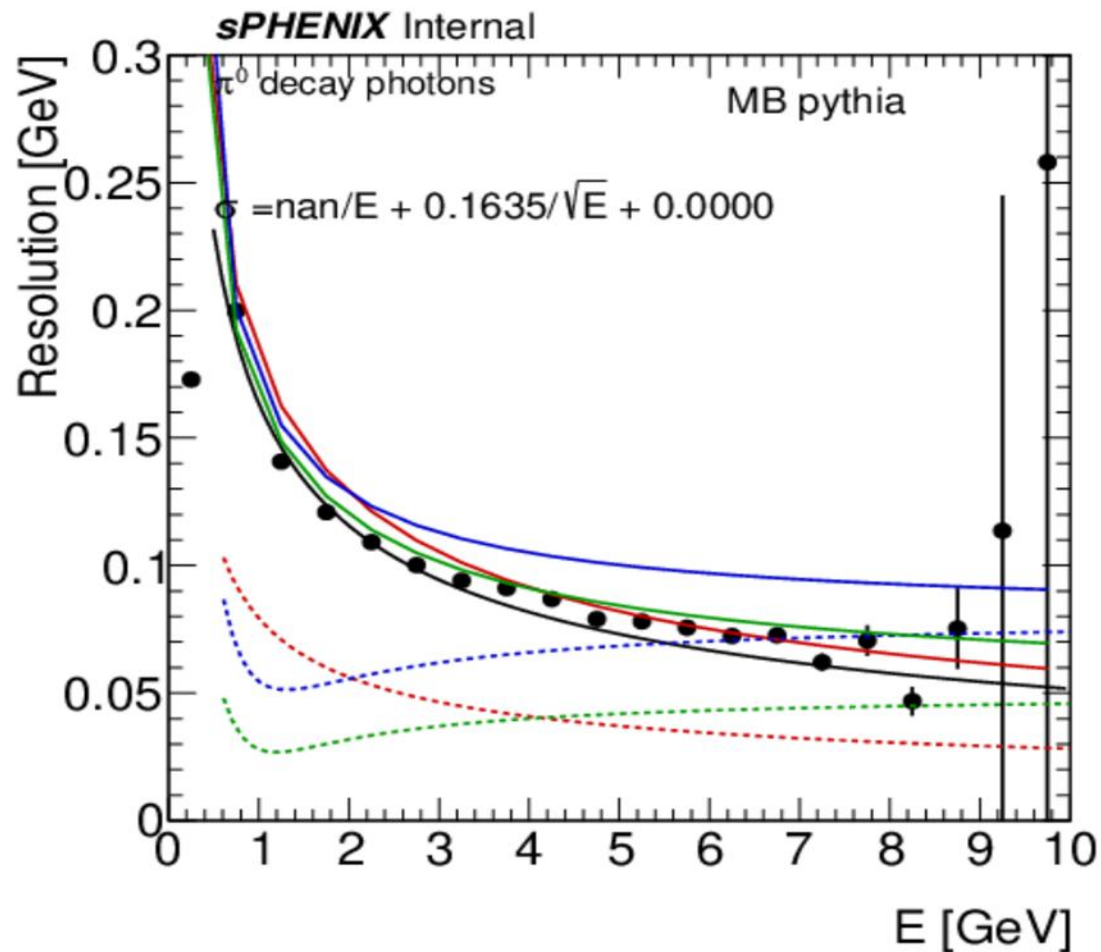
# Data-driven extraction of resolution

➤ [https://indico.bnl.gov/event/31859/contributions/120756/attachments/68548/117758/EMCal\\_toyMC\\_calib\\_20260302.pdf](https://indico.bnl.gov/event/31859/contributions/120756/attachments/68548/117758/EMCal_toyMC_calib_20260302.pdf)



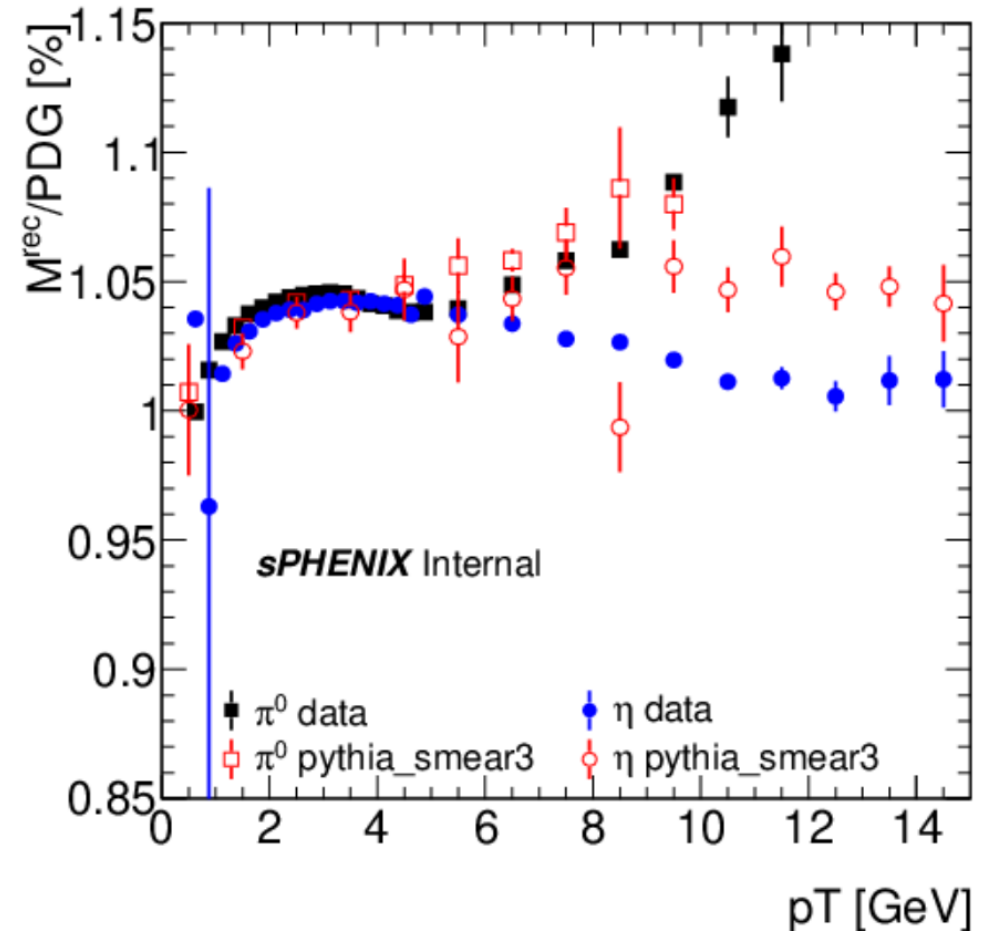
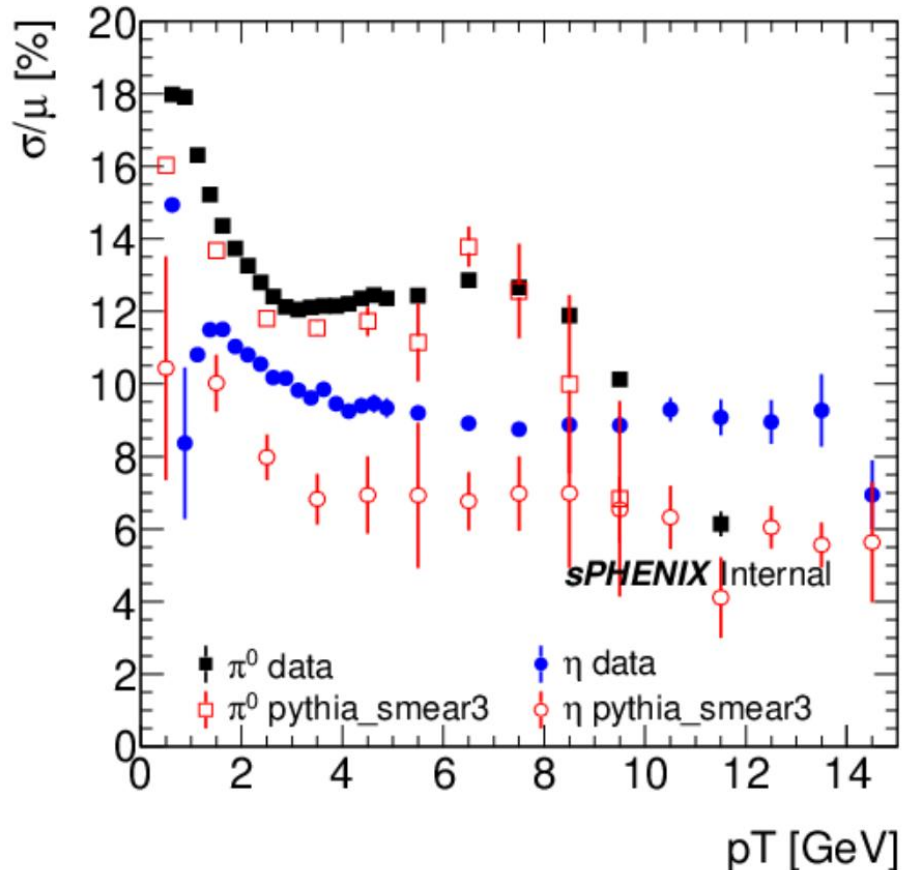
# Smearing factors

- Measure the energy and position resolution in MC to find how much additional smearing is needed to reproduce the data



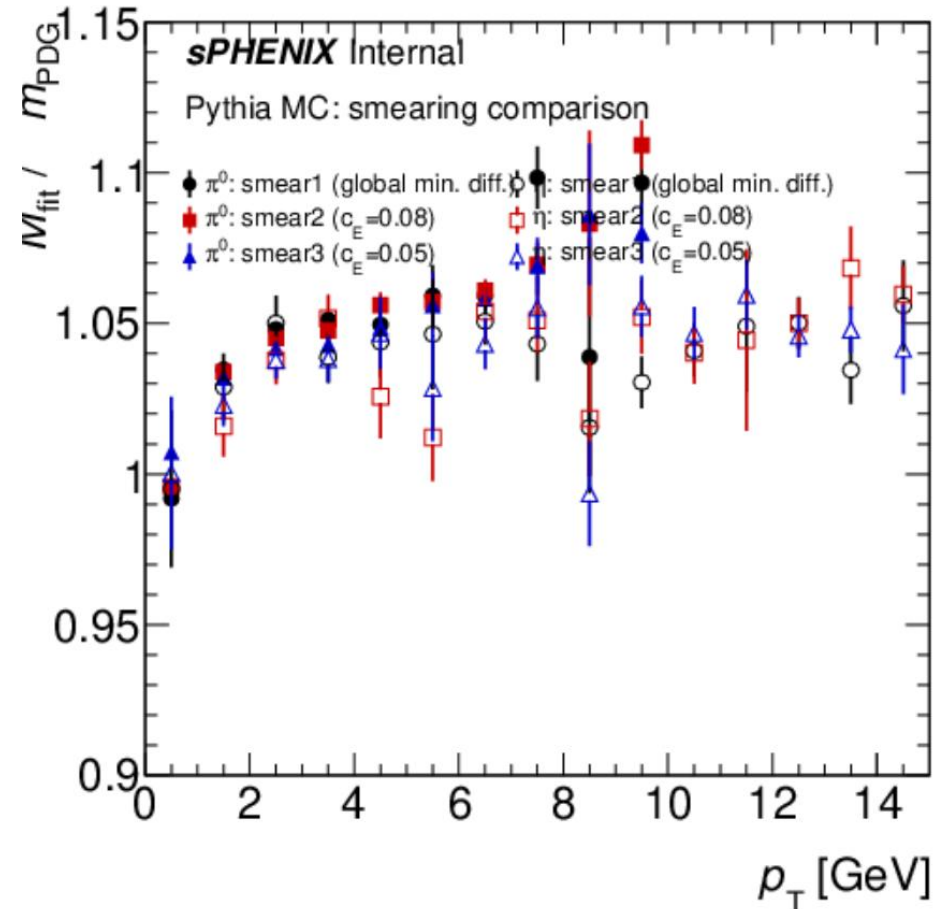
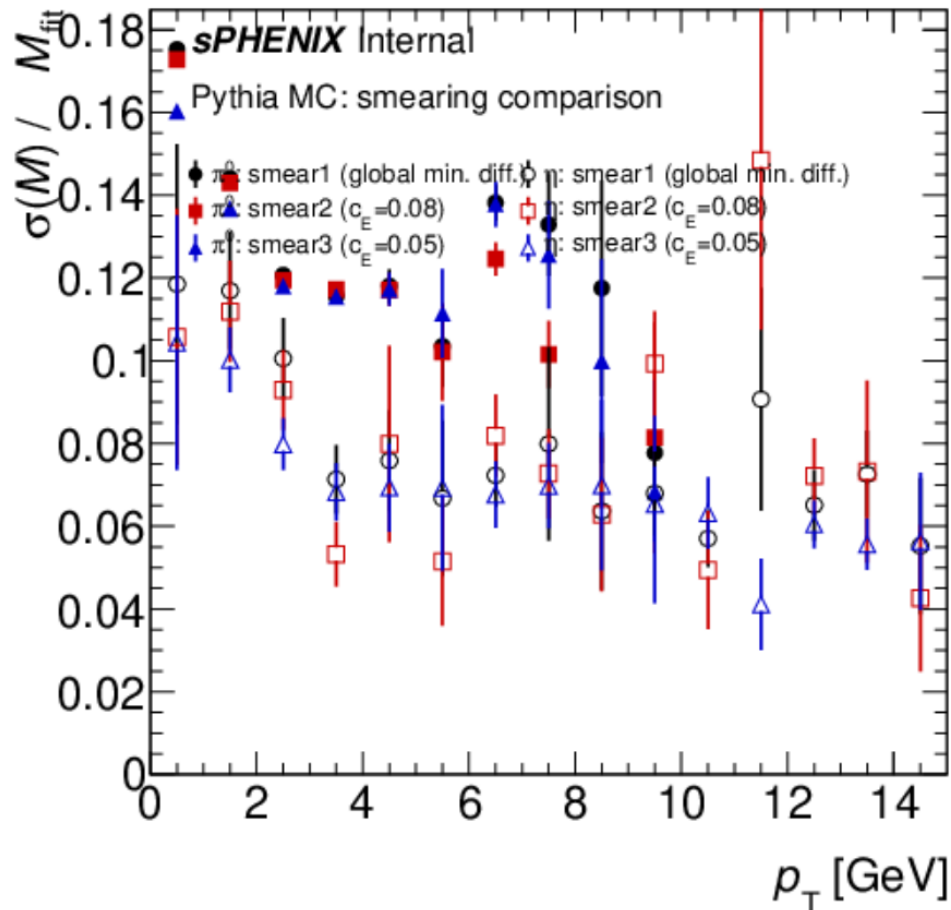
# Application of smearing

- Additional smearing gets the  $\pi^0$  to within about  $\sim 0.5\%$  which seem good, but the 3 variations (next slide) don't envelope data which would be preferable for systematics
- Eta is not well described and is a major issue
- Will try to addithal smearing to better describe the Eta



# Comparison of all the smearing sets

- Comparison of the 3 different additional smearings
- Although stats are not great, none of the 3 reproduce the data eta
- Little variation in the  $\pi^0$  mass and width.



# Conclusion

- **Nominal pythia has much better resolution than data**
- **It is unclear what is the major issue in data**
- **Extra smearing is helping describing the data but needs further tuning**
- **Need to also follow up on high  $p_T$   $\eta$  as it is the only constraint on calibration at high  $p_T$ .**